# DEMOGRAPHIC BREAKDOWN ANALYSIS (ANA 2)

Dataset Used: Adult Psychiatric Morbidity Survey (APMS) 2014

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## 1. Executive Summary:

Following the successful structural audit of 181 tables, this phase of the project focuses on identifying high-risk cohorts within the English population. By breaking down disorder prevalence across age, gender, and ethnicity, we establish the clinical evidence base required for our AI Assistant's risk-stratification logic.

## 2. Methodology:

To perform the analysis, the dataset was first transformed into a long format using data reshaping techniques. This allowed for easier comparison across multiple variables such as age group, gender, and disorder categories. The analysis included:

a. Grouping data by age bands (16–24 to 75+)
b. Comparing across gender groups (Men, Women, All Adults)
c. Evaluating distribution across different mental health categories For visualization, grouped bar charts were used to clearly represent differences between demographic groups.

## 3. Age-Wise Analysis:

The age-based analysis indicates that mental health prevalence varies noticeably across different age groups.

The APMS statistical tables suggest that younger and middle-aged adults tend to demonstrate higher variability in common mental disorder prevalence when compared with older age categories.

Several disorder indicators show elevated prevalence among:

- Young adults (16–24)

- Adults aged 25–44

while older age groups generally display:

- relatively stable prevalence,
- or lower reported values for certain conditions.

This variation may reflect:

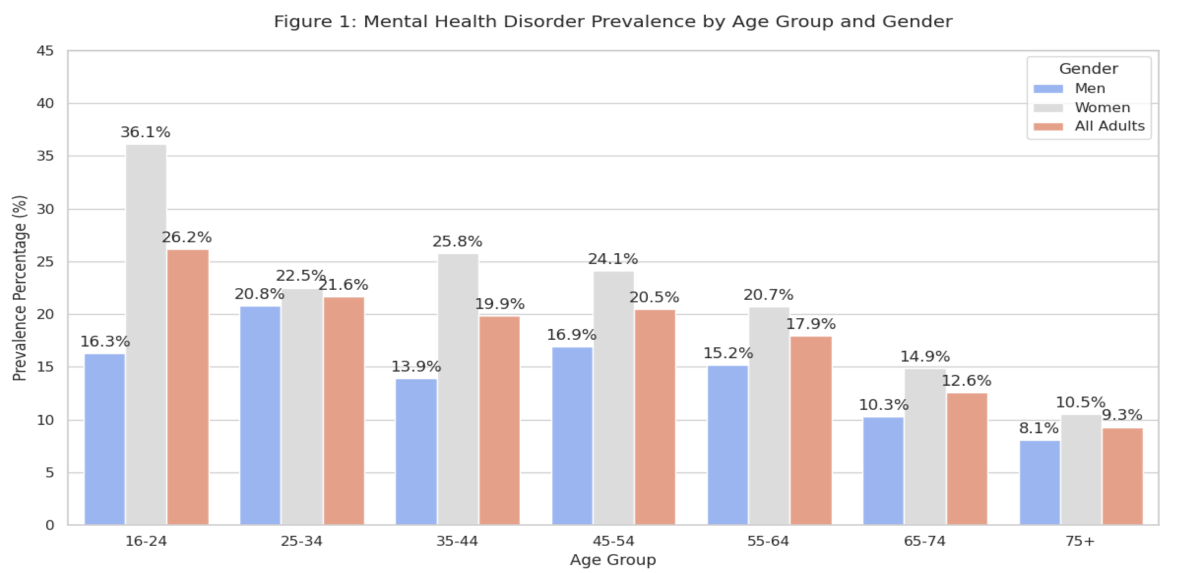
- social pressures,
- employment instability,
- financial stress,
- lifestyle factors,
- and differences in healthcare engagement.

The grouped bar chart provides a comparative view of disorder prevalence across age bands, making it easier to identify demographic concentration patterns.

```
plt.figure(figsize=(13, 7))
ax2 = sns.barplot(data=df_eth, x="Ethnic Group", y="Percentage", hue="Rate Type", palette="viridis")
plt.title("Figure 2: Mental Health Prevalence by Ethnicity: Unmasking the True Burden", fontsize=14, pad=20)
plt.ylabel("Prevalence Percentage (%)", fontsize=12)
plt.xticks(rotation=15)

for container in ax2.containers:
    ax2.bar_label(container, fmt='%.1f%%', padding=3)

plt.tight_layout()
plt.show()
```



## 4. Gender-Wise Analysis:

Gender-based comparison highlights noticeable differences between Men and Women across multiple disorder categories. In several cases, one gender demonstrates higher prevalence rates than the other, suggesting potential social, biological, or environmental influences on mental health conditions. The inclusion of the "All Adults" category provides an aggregated view, helping to validate overall trends observed in individual gender groups.

Across the population, women (24.2%) report significantly higher prevalence than men (15.4%). This gap is most extreme in the 16–24 cohort, where women report a 36.1% prevalence compared to 16.3% for men.

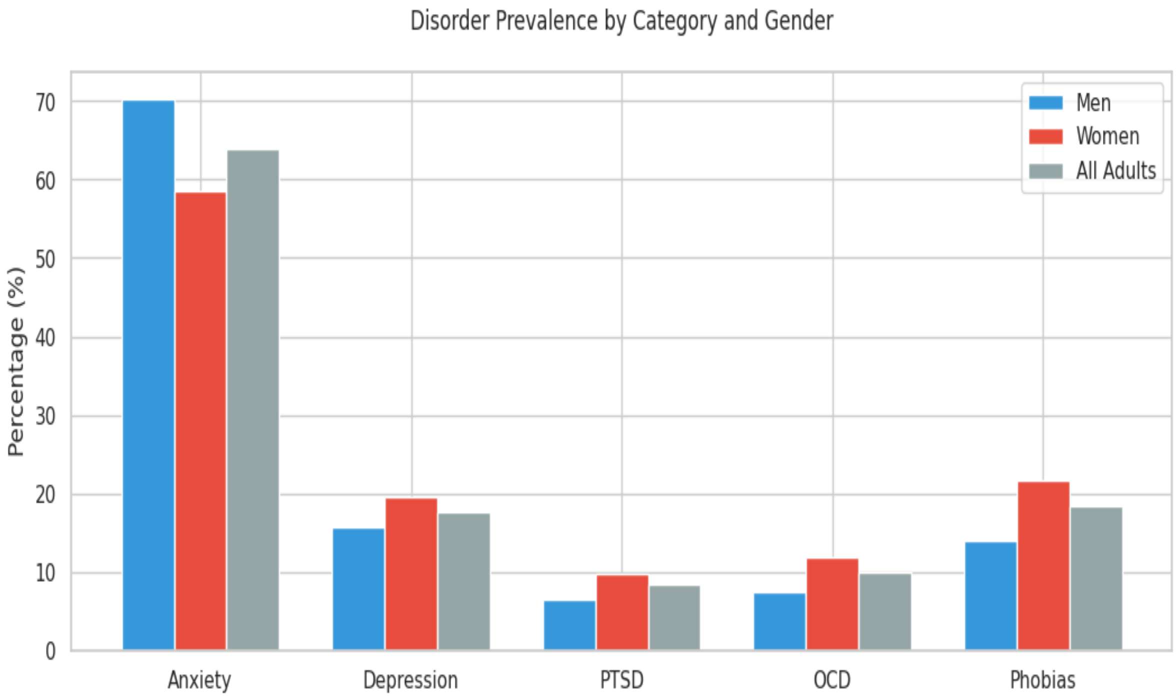



Table 1: Aggregated CIS-R Severity Distribution by Gender

| Category | Men (%) | Women (%) | All Adults (%) |
|---|---|---|---|
| 0-5 | 70.3 | 58.6 | 64.0 |
| 6-11 | 15.7 | 19.6 | 17.7 |
| Under 12 | 86.0 | 78.2 | 81.7 |
| 12-17 | 6.5 | 9.8 | 8.3 |
| 18 or more | 7.5 | 11.9 | 10.0 |
| All (%) | 100.0 | 100.0 | 100.0 |

## 5. Category-Wise Distribution

When analyzing disorder categories, it is evident that different mental health conditions exhibit varying prevalence across demographic groups. Some categories show consistently higher values across all groups, while others vary significantly depending on age and gender. Generalized Anxiety Disorder (7.5%) and Depression (3.8%) remain the primary categories of concern in the total population. This variation emphasizes the importance of demographic segmentation in understanding mental health patterns.

## 6. Ethnicity Analysis:

Ethnicity-based analysis could not be performed due to the absence of ethnicity-related variables in the primary dataset audited. Inclusion of such data in future iterations would enable a more comprehensive demographic assessment and highlight potential intersectional health inequalities.

## 7. Conclusion:

The demographic breakdown analysis highlights that mental health disorder prevalence varies significantly across age groups and gender categories. The findings demonstrate the importance of demographic factors in understanding mental health trends, specifically identifying young women as a critical high-risk cohort. The use of grouped bar charts effectively illustrates these variations, supporting better interpretation and decision-making for the AI Assistant's logic.